A Deep Temporal–Spectral–Spatial Anchor-Free Siamese Tracking Network for Hyperspectral Video Object Tracking

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***Abstract*— High spatial, high spectral, and high temporal (H^3) information of the objects of interest can be provided by hyperspectral video, which makes it possible to track objects in complex scenarios. However, during motion, changes in the target's appearance, background, and spectral information can degrade the performance of existing hyperspectral trackers due to insufficient training data. Consequently, this results in weak generalization of these trackers. In this article, to solve the above problems, a deep temporal–spectral–spatial anchor-free Siamese tracking network for hyperspectral video object tracking, namely HA-Net, is proposed. In HA-Net, a Siamese spectral enhancement tracker module based on an RGB tracker (pseudo-color tracker) is designed, which uses the powerful feature expression capabilities of the deep network to learn more discriminative deep spectral features for identifying objects in complex scenarios. The pseudo-color tracker is introduced to solve the problem of model performance limitation due to insufficient training data. By introducing the temporal–spectral–spatial online discrimination learning module, the temporal–spectral–spatial information of the target can be dynamically modeled to adapt to new targets and the dynamic changes of targets. Benefiting from the double Siamese network architecture, the model can be effectively trained from scratch with less than 20 000 training samples. Online learning of temporal–spectral–spatial information for the target, particularly in cases of insufficient training data, can alleviate the issue of model degradation. This approach enhances the model's robustness when tracking the target in complex scenes. In the 2021 IEEE WHISPERS Hyperspectral Object Tracking (HOT) Challenge, HA-Net obtained the best performance, with a distance precision (DP) score of 0.948 and an area under the curve (AUC) score of 0.688. The running speed is also 14 frames/s, which is superior to the existing hyperspectral object trackers for hyperspectral video. The source code is available at https://github.com/zhenliuzhenqi/HOT.**

***Index Terms*— Hyperspectral object tracking, online learning, Siamese network anchor-free, spectral classification.**

Received 2 March 2024; revised 1 June 2024 and 23 August 2024; accepted 17 September 2024. Date of publication 17 October 2024; date of current version 2 December 2024. This work was supported by the National Natural Science Foundation of China under Grant 42325105. *(Corresponding author: Yanfei Zhong.)*

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Digital Object Identifier 10.1109/TGRS.2024.3483072

I. Introduction

HYPERSPECTRAL video is a new type of data source that can provide high spatial, high spectral, and high temporal (H^3) information on the objects of interest. The aim of hyperspectral object tracker is to exploit hyperspectral video for object tracking. The region of the target of interest is manually annotated in the initial frame of the hyperspectral sequence, and then, the tracker predicts the region of the tracking object in the subsequent frames. This can also be regarded as one-shot learning [1]. Compared with RGB-based object tracking methods [2], [3], both spatial information and spectral information of tracking target can be exploited by hyperspectral video object tracking methods. This is especially the case for scenarios with background clutter (BC) and other complex attributes because the rich spectral information [4], [5] of the hyperspectral image (HSI) can be used to distinguish objects with similar appearances and different materials.

There are currently less than 20 000 training samples [6] and no pretrained models in the hyperspectral tracking community, which is insufficient training data [7], [8], [9] for deep learning-based hyperspectral trackers. In brief, when the training samples are not sufficient, deep learning-based methods do not perform as well as the traditional machine learning methods. Hence, the majority of contemporary hyperspectral trackers rely on pseudo-colored semantic information (i.e., pseudo-color image (PCI) is synthesized by selecting three bands from HSI as the input to the RGB-based tracker). Benefiting from pretrained models from ImageNet [10], [11] and tens of millions of RGB-based object tracking training samples [12], [13], [14], [15], RGB-based deep learning object trackers have been highly successful. The excellent performance of the object trackers [1], [16], [17], [18], [19] in recent years has also demonstrated that deep learning-based object trackers require a large number of training samples (millions or tens of millions of training data). The Siamese-based deep learning trackers are currently the most successful object tracking paradigms for RGB video. The current RGB-based deep Siamese trackers generally use an ImageNet pretrained model as the backbone and then use either LaSOT [13], TrackingNet [14], Got10K [12], COCO [20], ImageNet [11], or other datasets to train the deeper semantic layer of the backbone, the subsequent matching network, and prediction network. In other words, Siamese trackers require a large amount of training

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samples to learn a general feature representation to obtain a high-precision bounding box. However, due to the lack of a large amount of hyperspectral training samples, no pretrained models exist in the hyperspectral tracking community. Furthermore, due to the large amount of offline training samples required by the deep learning-based Siamese trackers [16], [21], the application of these methods in hyperspectral object tracking is limited.

In complex scenes, there will be changes in the target's appearance, background, and spectral information during motion, and new objects of interest and new scenes can degrade the existing tracking models for hyperspectral video and cause tracking drift. The CNHT [22], DeepHKCF [23], DMT [24], TASSCF [25], TSCFW [26], H^3Net [27], and MHT [6] methods based on a correlation filter [28], [29], [30], [31] that have been proposed in recent years use a linear weighting approach to update the model, and the model of the latest few frames occupies a large proportion. However, this linear weighting-based updating method cannot retain the information of historical frames, and it can easily result in error accumulation, which eventually leads to tracking drift. Li et al. [32] proposed BAE-Net and SST-Net [33], which generate multiple training samples according to the prediction results of each frame and apply stochastic gradient descent (SGD) iteration to optimize the generator and discriminator during the online update process [34]. However, the iterative updating method adopted by BAE-Net and SST-Net requires the updating of many parameters, and the calculation of the online optimization algorithm is complex, resulting in the model processing speed being less than 1 frames/s. Currently, state-of-the-art hyperspectral trackers, such as SiamHYPER [35], SEE-Net [36], SiamBAG [37], SENSE [38], and TransHST [39], exhibit notable deficiencies in their online update capabilities. This limitation renders these trackers vulnerable to model degradation, resulting in tracking drift, particularly in complex scenarios with limited training data. At the same time, single-object tracking is similar to a one-shot task, and after the tracking object is manually annotated in the first frame, it needs to be detected in subsequent frames [1]. Therefore, there is a need for a deep hyperspectral tracker with swift online optimization capabilities to counteract model degradation resulting from changes in target appearance, background, and spectrum. This approach aims to mitigate the risk of overfitting, particularly in scenarios with limited training data.

In this article, to solve the above problems, a deep temporal–spectral–spatial anchor-free Siamese tracking network for hyperspectral video object tracking, namely, HA-Net, is proposed. In HA-Net, the double Siamese network structure is designed to solve the problem of model performance limitation due to insufficient training data. Training a pseudo-color tracker with massive data (RGB data) provides a strong baseline, and integrating hyperspectral features on this foundation can yield exceptional tracking results. By adding a Siamese spectral enhancement tracker module (the Siamese spectral enhancement tracker module is trained from scratch) to the pseudo-color tracker, the deep hyperspectral semantic information is used to enhance the discrimination ability of the pseudo-color-based deep Siamese tracker. The temporal–spectral–spatial online discrimination learning module is used to model the temporal–spectral–spatial information. The online discrimination learning module retains the initial hyperspectral template to prevent error accumulation, while the dynamic hyperspectral template is added to accommodate changes in target information during motion. Especially in the absence of training data, the parameters of the online discrimination learning module are quickly optimized based on the target template given in the first frame, allowing the model to adapt to new targets (similar to one-shot learning for detection tasks [1]). The hyperspectral anchor-free prediction module can directly predict the ratio and size of the BBOX of the target, avoiding the hyperparameters introduced by the predefined bounding boxes, making the prediction of the BBOX more consistent with the changes in the appearance of the target, and more robust.

The main contributions of this article are as follows.

1) HA-Net can extract highly discriminative semantic features even with limited datasets, significantly enhancing the model's generalization. In addition, the pseudo-color tracker is introduced to address the performance limitations caused by insufficient training data.
2) The temporal–spectral–spatial online discrimination learning module is proposed to model the temporal–spectral–spatial information during the tracking process through a fast online optimization algorithm, to adapt to the dynamic changes (changes in appearance, spectrum, background, and so on) of the target and background. The initial hyperspectral template is used to prevent error accumulation, while the dynamic hyperspectral template is added to accommodate changes in target information during motion.
3) Extensive experiments have substantiated the efficacy of HA-Net. Notably, HA-Net, when trained with a smaller dataset than that offered by the WHISPERS dataset, attains outstanding performance. This underscores the online learning capability of HA-Net, especially in situations involving limitations in training data.

HA-Net achieved first place in the 2021 IEEE Hyperspectral Object Tracking Challenge.[1] At the same time, the proposed tracker can run at 14 frames/s on a graphics processing unit (GPU), far exceeding the speed of the previous hyperspectral trackers.

The rest of this article is organized as follows. Section II introduces the related work in RGB video object tracking and hyperspectral video object tracking. Section III gives the details of the proposed hyperspectral object tracking framework. Section IV provides the experimental results and experimental analysis. Section V provides the ablation experiment results. Section VI provides the data scale ablation experiment results. Section VII concludes this article.

II. Related Work

The deep hyperspectral Siamese network for hyperspectral video proposed in this article is related to two aspects of

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work: 1) RGB video object tracking and 2) hyperspectral video object tracking.

A. RGB Video Object Tracking

Bertinetto et al. [21] proposed SiamFC, which uses massive RGB training datasets to train a Siamese network-based model offline and then uses cross correlation to calculate the response map. Dong et al. [40] proposed the integrated circulant structure kernels (ICSKs) method and obtained good results by using a novel two-stage classifier with a circulant structure with kernels to solve the problem of heavy occlusions. Dong et al. [41] first proposed dynamic hyperparameter optimization in a reinforcement learning framework for object tracking. In 2018, the SiamRPN method proposed by Li et al. [1] was the first method to use a region proposal network (RPN) [42] for object tracking. In 2019, SiamRPN++ was proposed by Li et al. [16] to solve the issue of the Siamese trackers that can only use a shallow network to extract the features of the object. At the same time, a depth-wise cross correlation (depth-cross) layer [16] was introduced to solve the issues of a large number of parameters and unstable network training. Subsequently, a large number of trackers have been developed based on SiamRPN++, including SiamFC++ [43], the Siamese box adaptive network (SiamBAN) [44], SiamAttn [45], SiamCAR [17], and Ocean [18].

With the successful integration of the self-attention mechanism [46] in vision, an increasing number of trackers are leveraging this mechanism, leading to enhanced performance. Wang et al. [47] employed a transformer to model temporal contextual relationships among multiple frames, thereby facilitating the generation of high-quality tracking models. Chen et al. [48] proposed TransT-M, utilizing transformer to replace the cross correlation layer [16] and thereby preventing the loss of semantic information within the cross correlation layer. Cui et al. [49] introduced MixFormer, proposing the extraction and fusion of features through a mixed attention module to fully leverage self-attention. Ye et al. [50] proposed OSTrack, which uses an early candidate elimination module to gradually discard candidates belonging to the background, avoiding the negative effects of uninformative background regions.

Evidently, current state-of-the-art trackers are predominantly either Siamese-based or transformer-based. However, offline-trained models are prone to degradation, especially in the face of complex scenarios with limited training samples.

B. Hyperspectral Video Object Tracking

Uzkent et al. [23], [51], [52], [53] proposed a series of hyperspectral object tracking algorithms for aerial hyperspectral video, using features such as hyperspectral feature-based blob extraction [51], features based on local hyperspectral histograms [52], hyperspectral likelihood maps [53], and pseudo-color features from deep learning (where PCI is synthesized by selecting three bands from HSI, which is then input into the model trained with an RGB dataset) [23]. Qian et al. [22] proposed the CNHT method based on the fixed convolution kernels, where the convolution kernels are randomly selected in the initialized target area, and the convolution kernels do not need to be trained. The histogram of oriented gradients (HOGs) [54] features is replaced by the extracted hyperspectral features for input into the KCF [28] tracker. MHT tracker proposed based on BACF [29] tracker by Xiong et al. [6]. The SSHMG and abundances of the material compositions of objects in the MHT tracker can simultaneously provide spatial and spectral information, making the MHT tracker achieve very good results. The BAE-Net and SST-Net [33] proposed by Li et al. [32] use a band combination module to compose PCIs synthesized in different bands for input into the visual tracking via adversarial learning (VITAL) [34] tracker. Tang et al. [55] introduced BAHT, incorporating a background-aware band selection module into the Siamese tracker for hyperspectral tracking. Liu et al. [35] proposed SiamHYPER to solve the data-hungry problem of hyperspectral object tracking community using PCI and HSI fusion (dual Siamese networks structure) to obtain excellent results. Subsequently, Tang et al. [56] proposed SiamHT to use a dual Siamese network structure for hyperspectral tracking. Li et al. [36] proposed SEE-Net, which employs a spectral self-expressive module to learn crucial hyperspectral bands for hyperspectral video tracking.

The existing advanced hyperspectral object trackers face three main challenges. First, the first challenge is that the deep hyperspectral trackers [23], [32], [35], [36], [56] face a significant risk of overfitting, leading to weak generalizability of these trackers. Second, the second challenge involves changes in the target during tracking within complex scenes, which can result in the degradation of the tracking model and the occurrence of tracking drift. Third, the final challenge is associated with the current online update strategies employed by hyperspectral object trackers, which tend to be overly complex and nonrobust. The complex update strategy of the VITAL-based tracker, which requires fine-tuning for each frame, results in a slow running speed for the VITAL-based hyperspectral tracker. On the other hand, the linear update strategy tends to accumulate errors over time, eventually causing tracking drift.

The temporal–spectral–spatial online discrimination learning (TSS) module for HA-Net has two main functions: 1) utilizing the first frame of the video to learn the strong discriminative features of the target online and adaptively updating the hyperspectral dynamic template with the classification scores and 2) retaining the initial template and fusing the results of the initial template with the dynamic template. Compared with the VITAL-based update method that requires adversarial learning for each frame, the TSS module utilizes only the first frame for online learning, which improves the inference speed. Traditional weighting-based update methods are prone to accumulating errors, which eventually lead to tracking drift. The TSS module is able to perform online learning of strongly discriminative features while retaining the initial template, which avoids the accumulation of errors. This is particularly beneficial for hyperspectral tracking, where training data are insufficient, as online learning enables the model to have strong generalization ability.

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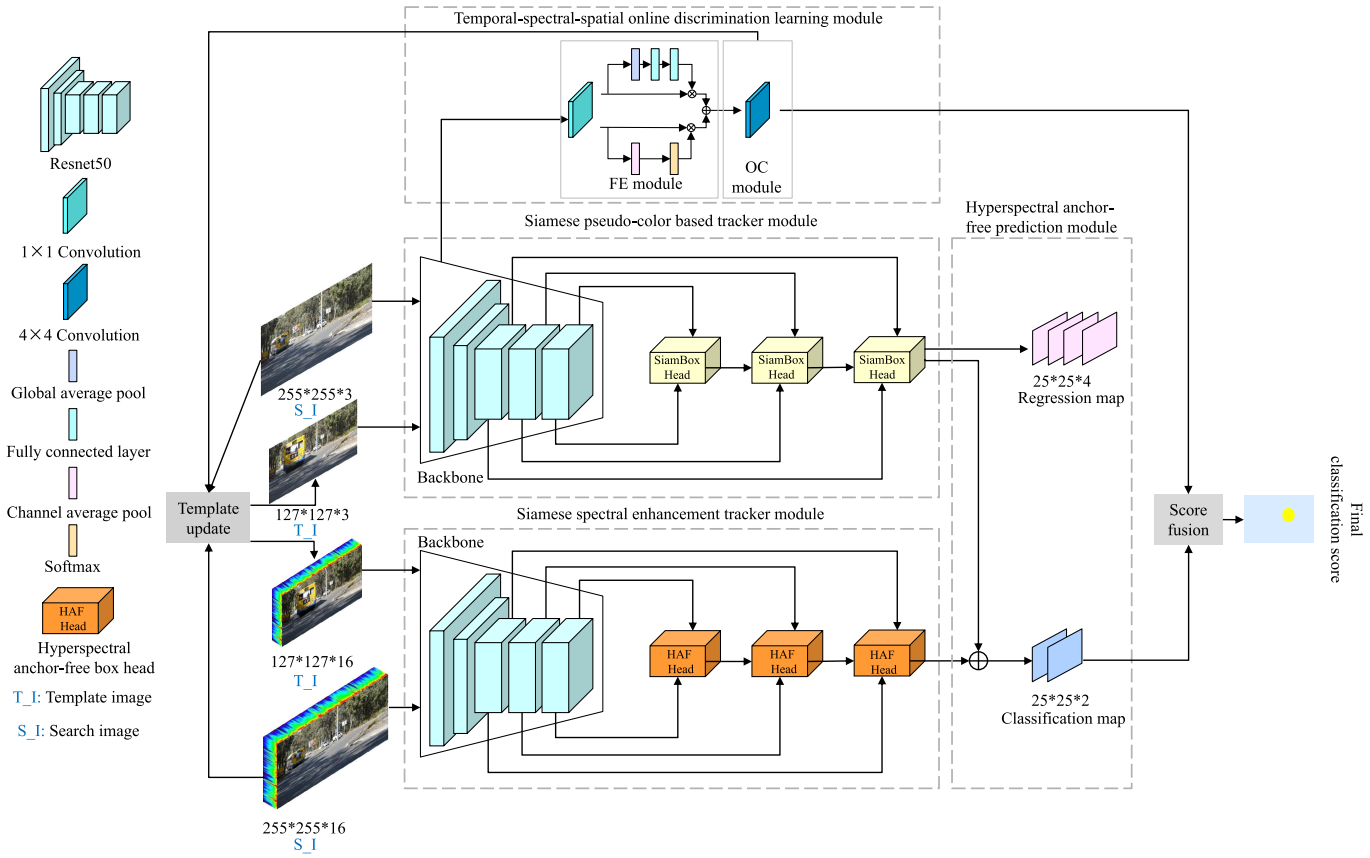


Fig. 1. Proposed HA-Net framework. The main components of the framework are the double Siamese network (Siamese spectral enhancement tracker module + Siamese pseudo-color-based tracker module), the temporal–spectral–spatial online discrimination learning module, and the hyperspectral anchor-free prediction module. The Siamese spectral enhancement tracker module takes an HSI as input. The pseudo-color-based Siamese tracking network takes a PCI synthesized from three bands in the HSI as input. The temporal–spectral–spatial online discrimination learning module is responsible for updating the hyperspectral template and the corresponding pseudo-color template.

III. Deep Temporal–Spectral–Spatial Anchor-Free Siamese Tracking Network for Hyperspectral Video Object Tracking

As shown in Fig. 1, HA-Net can be divided into four main parts: 1) the Siamese pseudo-color-based tracker module; 2) the Siamese spectral enhancement tracker module; 3) the hyperspectral anchor-free prediction module; and 4) the temporal–spectral–spatial online discrimination learning module. The Siamese spectral enhancement tracker module is responsible for extracting multilevel deep spectral features based on the Siamese network. A hyperspectral enhancement classification feature map and a hyperspectral enhancement regression feature map are the outputs of the Siamese spectral enhancement tracker module. The Siamese pseudo-color-based tracker module is SiamBAN (PSE-SiamBAN tracker module), using the off-the-shelf model. A regression map and a classification map are the outputs of PSE-SiamBAN tracker module. The hyperspectral anchor-free prediction module is responsible for fusing the two classification maps and uses the fused map to predict the object category. The pseudo-color-based Siamese tracker, based on massive training samples, can solve the problem of insufficient hyperspectral video training data. The addition of a Siamese spectral enhancement tracker module, using hyperspectral semantic features, can effectively increase the discriminability of the PSE-SiamBAN tracker module. On the other hand, the hyperspectral anchor-free prediction module uses the regression map output from PSE-SiamBAN to calculate the BBOX of the tracking object. The temporal–spectral–spatial online discrimination learning module is responsible for the online learning and update of the hyperspectral template, to solve the one-shot task and adapt to the changes of the foreground and background during movement. Especially in scenarios with a lack of training data, online learning empowers HA-Net to achieve robust generalization.

A. Siamese Pseudo-Color Based Tracking Network

After Li et al. [1] introduced the RPN to Siamese-based tracking, Siamese trackers based on deep learning have achieved excellent real-time running speeds. SiamRPN++ that solved the issue of the Siamese-based object trackers can only use shallow networks by changing the training method. Siamese trackers have since made considerable progress. In the past two years, Siamese trackers [16], [17], [43], [44] have divided the scale and position of the predicted target into two parts: classification and regression. Using the inherent properties of the RGB-based deep Siamese trackers, a Siamese spectral enhancement tracker module can be added when the hyperspectral training samples are limited, and the

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hyperspectral multilevel features can be used to predict the category of the target.

In this study, we chose SiamBAN as the pseudo-color-based Siamese tracking module. We used the off-the-shelf model of SiamBAN for joint training and inference. Unlike RGB-based object tracking, three bands in the HSIs are used to synthesize three-band PCIs as the input of PSE-SiamBAN. The input of the pseudo-color template (pse-template) branch of PSE-SiamBAN is a three-band pseudo-color template image denoted as $z_{3\text{bands}}$, and the input of the pseudo-color search (pse-search) branch of PSE-SiamBAN is a three-band pseudo-color search image denoted as $x_{3\text{bands}}$. ResNet-50 is adopted by PSE-SiamBAN as the backbone network to extract pseudo-color features. The feature extracted by the pse-template branch is marked as $\varphi(z_{3\text{bands}})$, and the feature extracted by the pse-search branch is marked as $\varphi(x_{3\text{bands}})$. These features are used to generate a pseudo-color classification feature (*pse-class*) map and pseudo-color regression feature (*pse-reg*) map through SiamBox heads. Similarly, the *pse-class* map and *pse-reg* map can be obtained as shown in the following equations:

$$P^{\text{cls}}_{w*h*2} = [\varphi(z_{3\text{bands}})]^{\text{cls}} \star [\varphi(x_{3\text{bands}})]^{\text{cls}} \tag{1}$$

$$P^{\text{reg}}_{w*h*4} = [\varphi(z_{3\text{bands}})]^{\text{reg}} \star [\varphi(x_{3\text{bands}})]^{\text{reg}} \tag{2}$$

where P^{cls}_{w*h*2} is the *pse-class* map, P^{reg}_{w*h*4} is the *pse-reg* map, and $\star$ is the depth-cross operation. The *pse-class* maps and the *pse-reg* maps generated by the features of the *convolution3* layer, *convolution4* layer, and *convolution5* layer of PSE-SiamBAN's are adaptively fused as follows:

$$P^{\text{cls_all}}_{w*h*2} = \sum_{n=3}^{5} \zeta_n P^{\text{cls}}_n \tag{3}$$

$$P^{\text{reg_all}}_{w*h*4} = \sum_{n=3}^{5} \eta_n P^{\text{reg}}_n \tag{4}$$

where ζ_n and η_n, respectively, correspond to the weights of the *pse-class* map and the *pse-reg* map, which are automatically optimized by the network.

B. Siamese Spectral Enhancement Tracker Module

Due to a shortage of training data, contemporary hyperspectral object trackers primarily rely on pseudo-color features. To solve this problem, we propose a double Siamese network architecture. Based on the pseudo-color Siamese network, a Siamese spectral enhancement tracker module is added for extracting hyperspectral semantic information. The hyperspectral semantic information is used to improve the classification results based on the pseudo-color-based Siamese network, to improve the prediction accuracy of the whole model. The Siamese spectral enhancement tracker module we designed also adopts a Siamese network structure and uses the *depth-cross* layer contained in the hyperspectral anchor-free box head to output the hyperspectral classification enhancement feature (*hyper-class*) map and the hyperspectral regression enhancement feature (*hyper-reg*) map, as shown in Fig. 2. The assumption is that during training, the classification map and regression map can promote each other and obtain better results.

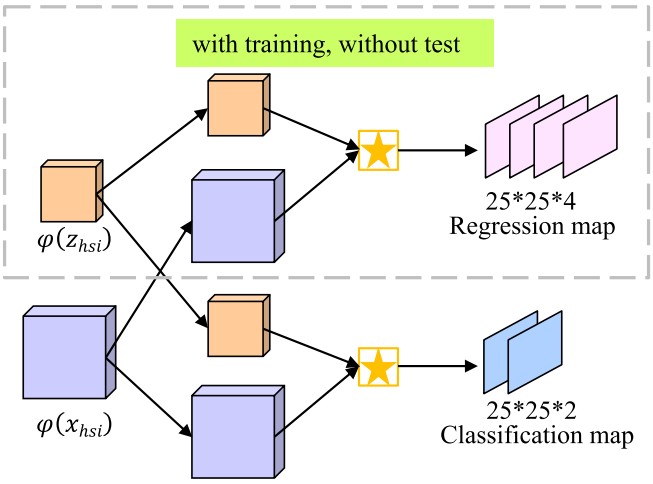


Fig. 2. Hyperspectral anchor-free box head. When training, the classification enhancement branch and the regression enhancement branch are trained simultaneously. When testing, the regression enhancement branch is removed, leaving only the classification enhancement branch.

The input of the Siamese spectral enhancement tracker module is HSIs (with all the hyperspectral bands). The Siamese spectral enhancement tracker module is divided into two branches, one of which is the hyperspectral template (hyper-template) enhancement branch and the other is the hyperspectral search (hyper-search) enhancement branch. The hyper-template image z_{hsi} is used as the input of the hyper-template branch, and the hyper-search image x_{hsi} is used as the input of the hyper-search branch. The input channels of *convolution1* layer of ResNet-50 were modified from 3 to 16. The remaining structure of ResNet-50 remained unchanged, enabling it to accommodate 16 channels of hyperspectral data as input. ResNet-50, from which the downsampling operation of the last two convolutional blocks has been removed, is used to extract hyperspectral features for the dense object tracking prediction. Different dilated rates for the dilated convolution can obtain different object scales, so the dilated rate of the *convolution4* layer was set to 2 and the dilated rate of the *convolution5* layer to was set 4. The stride of the *convolution4* layer and *convolution5* layer was set to 1.

The shallow features are responsible for the location accuracy of the tracker, while the semantic features are responsible for the robustness of the tracker. Therefore, the shallow hyperspectral features and the semantic hyperspectral features are fused to predict the category of the hyperspectral targets. The features of the *convolution3* layer, *convolution4* layer, and *convolution5* layer of ResNet-50 are the outputs for the prediction of the hyperspectral target's category. In the case of limited hyperspectral training samples, the proposed framework can use multilevel deep spectral semantic features to predict the target's category. A 1×1 convolution is added to change the output feature channels of the *convolution3* layer, *convolution4* layer, and *convolution5* layer to 256, to reduce the computational burden. The output feature of the hyper-template enhancement branch is $\varphi(z_{\text{hsi}})$, and the output feature of the hyper-search enhancement branch is $\varphi(x_{\text{hsi}})$. As shown in Fig. 2, the output of the hyperspectral anchor-free

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box head is divided into a *hyper-class* map and a *hyper-reg* map. The *hyper-class* map and *hyper-reg* map are obtained through the following equations:

$$H^{\text{cls}}_{w*h*2} = [\varphi(z_{\text{hsi}})]^{\text{cls}} \star [\varphi(x_{\text{hsi}})]^{\text{cls}} \tag{5}$$

$$H^{\text{reg}}_{w*h*4} = [\varphi(z_{\text{hsi}})]^{\text{reg}} \star [\varphi(x_{\text{hsi}})]^{\text{reg}} \tag{6}$$

where H^{cls}_{w*h*2} is the *hyper-class* map, H^{reg}_{w*h*4} is the *hyper-reg* map, and $\star$ is the *depth-cross* operation. The features output by the three blocks are used as the input of the three hyperspectral anchor-free box heads to generate the *hyper-class* maps and *hyper-reg* maps for multilevel feature fusion, to predict the object's state. Equations (7) and (8) are used to obtain the fused map from the *hyper-class* maps and *hyper-reg* maps

$$H^{\text{cls_all}}_{w*h*2} = \sum_{n=3}^{5} \alpha_n H^{\text{cls}}_n \tag{7}$$

$$H^{\text{reg_all}}_{w*h*4} = \sum_{n=3}^{5} \beta_n H^{\text{reg}}_n \tag{8}$$

where α_n and β_n, respectively, correspond to the weights of the *hyper-class* map and the *hyper-reg* map, which are automatically optimized by the network.

C. Hyperspectral Anchor-Free Prediction Module

1) Joint Test Phase: In the test phase, *hyper-reg* map of the Siamese spectral enhancement tracker module is removed, leaving only the *hyper-class* map of the Siamese spectral enhancement tracker module. The *pse-class* map $P^{\text{cls_all}}_{w*h*2}$ and *hyper-class* map $H^{\text{cls_all}}_{w*h*2}$ are fused as follows:

$$HP^{\text{cls}}_{w*h*2} = \lambda_1 H^{\text{cls_all}}_{w*h*2} + \lambda_2 P^{\text{cls_all}}_{w*h*2} \tag{9}$$

where λ_1 and λ_2, respectively, correspond to the weights of the *hyper-class* map and the *pse-class* map, which are automatically optimized by the network. $\text{HP}^{\text{cls}}_{w*h*2}$ is the final classification feature map, which is used to distinguish the foreground from the background. The BBOX of the tracking object is obtained through the *pse-reg* map $P^{\text{reg_all}}_{w*h*4}$. By adopting the anchor-free prediction method, the inference speed of the model is further improved. Each point (i, j) of $P^{\text{reg_all}}_{w*h*4}$ contains a 4-D vector $d_p(i, j) = (l_p, t_p, r_p, b_p)$, which represents the distance from the corresponding position of the search area to the four sides of the BBOX. Therefore, the predicted bounding box can be obtained through the following equation:

$$\begin{aligned} b_{x1} &= p_x - l_p \\ b_{y1} &= p_y - t_p \\ b_{x2} &= p_x + r_p \\ b_{y2} &= p_y + b_p \end{aligned} \tag{10}$$

where (b_{x1}, b_{y1}), and (b_{x2}, b_{y2}), respectively, represent the coordinates of the top-left corner and the bottom-right corner of the predicted BBOX, and (p_x, p_y) represents the position corresponding to point (i, j) in $P^{\text{reg_all}}_{w*h*4}$.

2) Joint Training Phase: In the joint training process, the classification map HP^{cls}_{w*h*2} obtained by (9) is used to calculate the classification loss, and the hyperspectral regression map $H^{\text{reg_all}}_{w*h*4}$ obtained by (8) is adopted to calculate the regression loss. We believe that the simultaneous training of the regression branch network and the classification branch network can effectively promote the effects of the two branches. The distance from position (h_x, h_y) to the four sides of the ground truth can be obtained through the following equation:

$$\begin{aligned} l_h &= h_x - g_{x1} \\ t_h &= h_y - g_{y1} \\ r_h &= g_{x2} - h_x \\ b_h &= g_{y2} - h_y \end{aligned} \tag{11}$$

where l_h, t_h, r_h, and b_h are the distances from position (h_x, h_y) to the four sides of the BBOX and (h_x, h_y) represents the position corresponding to point (i, j) in $H^{\text{reg_all}}_{w*h*4}$. (g_{x1}, g_{y1}) and (g_{x2}, g_{y2}), respectively, represent the coordinates of the top-left corner and the bottom-right corner of the ground-truth bounding box. A cross-entropy loss is used as the classification loss, and an intersection over union (IoU) loss [57] is used as the regression loss. The total loss is as follows:

$$L = \varphi_1 L_{\text{cls}} + \varphi_2 L_{\text{reg}} \tag{12}$$

where L_{cls} denotes the cross-entropy loss and L_{reg} denotes the IoU loss. In this study, we set $\varphi_1 = 2$ and $\varphi_2 = 1$. The IoU loss is defined as

$$L_{\text{IoU}} = 1 - \text{IoU} \tag{13}$$

where IoU means the area proportion of the IoU of the BBOX calculated by $H^{\text{reg_all}}_{w*h*4}$ and the ground-truth BBOX.

D. Temporal–Spectral–Spatial Online Discrimination Learning Module

Especially when facing limitations in training data, dynamic changes in targets within complex scenes can result in model degradation, such as spectral changes and appearance changes. Single-object tracking aims to manually select any object of interest in the first frame and predict the state of that object in subsequent frames. Single-object tracking can be considered a one-shot task [1]. The risk of overfitting the hyperspectral tracking model is enhanced when unknown scene targets appear. The existing hyperspectral trackers [6], [22], [23], [32] do not solve this problem well and cannot model the changed appearance and spectrum of the target well during the movement of the target. To address this issue, inspired by ATOM [58], and DROL [59], a temporal–spectral–spatial online discrimination learning module is proposed. The temporal–spectral–spatial online discrimination learning module proposed in this article differs from the online module of the ATOM tracker in three ways.

1) Based on the online module of ATOM, we add a spatial attention layer and a channel attention layer. We use the first frame to fine-tune the parameters of the attention layers.

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2) We introduce a template updating strategy by updating the dynamic template to adapt to the dynamic changes of the target.
3) One-shot learning method is being applied for the first time in the field of hyperspectral tracking. The online learning module can improve the accuracy and robustness of the model in cases of insufficient training data.

Unlike the double Siamese network architecture that requires offline training, the temporal–spectral–spatial online discrimination learning module uses online learning to predict the classification score of the target. The online update process is divided into two steps: 1) solving the model degradation by fine-tuning the feature extraction (FE) module online using a manually annotated first frame and 2) updating the hyperspectral template online according to the online classification (OC) module, which can adapt to changes in the target and background. The temporal–spectral–spatial online discrimination learning module selects a high-quality hyperspectral template z_{hsi}^{d} by calculating the OC score of the target. The hyperspectral template z_{hsi}^{d} represents the dynamic hyperspectral template, which is different from the fixed first-frame template of the Siamese tracker. The dynamic hyperspectral template z_{hsi}^{d} is updated with the shift of the target's spectrum and appearance, to enhance the discriminant of HA-Net. In the process of online learning, the OC loss is defined as

$$L(w) = \sum_{j=1}^{m} \gamma_j \| f\left(x_j; w\right) - y_j \|^2 + \sum_{k} \delta_k \| w_k \|^2 \tag{14}$$

where $r_j(w) = (\gamma_j)^{1/2}(f(x_j; w) - y_j)$, $r_{m+k}(w) = (\delta_k)^{1/2} w_j$ is a residual problem, and $y_j \in R^{W \times H}$ is the annotated label, which is a Gaussian function centered at the tracking target's bounding box. The amount of regularization on w_k is set by δ_k. Equation (14) can then be denoted as $L(w) = \|r(w)\|^2$, which is the squared L^2 norm of the residual vector. This is a positive definite quadratic problem, which is solved by the optimization strategy proposed by Danelljan et al. [58], i.e., conjugate gradient descent is adopted to solve the quadratic problem. After selecting the target of interest, the FE module is fine-tuned by online learning to optimize the parameters for that particular target.

The structure of the temporal–spectral–spatial online discrimination learning module consists of two parts: 1) an FE module and 2) an OC module. The input of the temporal–spectral–spatial online discrimination learning module is the pseudo-color features output from the backbone network of SiamBAN, as shown in Fig. 1. The FE module is composed of a dimensionality reduction module and an attention module. In the dimensionality reduction module, 1×1 convolution and a rectified linear unit (ReLU) activation function are used to reduce the feature dimension to 64. In the attention module, spatial attention and channel attention are used to aggregate the features. The spatial attention module includes global average pooling (GAP) and two fully connected (FC) layers, and the channel attention module includes channel average pooling and softmax layers. In the OC module, a 4×4 kernel is used in the last layer to obtain a single-channel output. The data augmentation method is used to expand the first frame into 30 initial training samples. These 30 training samples are then used to fine-tune the parameters of the FE module. The features obtained by the FE module are used by the OC module for OC and scoring in subsequent frames. The hyperspectral classification score (donated as s_{online}) obtained by the OC module is weighted and fused with the classification score (donated as s_{hp}, which is calculated by HP_{w*h*2}^{cls} obtained by the double Siamese network, to obtain the final classification score, which can be formulated as

$$s = \theta s_{\text{online}} + (1 - \theta) s_{\text{hp}} \tag{15}$$

where θ represents the weight of the fusion of the two classification scores, which was set to 0.8 in this study. Cubic interpolation is used to interpolate the online hyperspectral classification score to the same spatial size as the classification score output by the double Siamese network.

The quality of the template of the HA-Net has an important influence on the performance of the tracker. In order to solve the problem of model degradation due to object changes during the tracking, an online update strategy is proposed for the dynamic hyperspectral template z_{hsi}^{d}. A high-quality dynamic hyperspectral template is selected through the maximum hyperspectral OC score (donated as $s_{\text{online}}^{\max}$), which can be formulated as

$$z_{\text{hsi}}^{d} = \boldsymbol{correspondingz}\left(s_{\text{online}}^{\max}\right)_{s_{\text{online}}^{\max} > \tau} \tag{16}$$

where ***correspondingz*** represents the hyperspectral template corresponding to the hyperspectral OC score, and τ represents the threshold used to suppress low-quality hyperspectral templates, which was set to 0.75 in this study. When the following equation is satisfied, the hyperspectral template is updated:

$$z_{\text{hsi}}^{new} = \begin{cases} z_{\text{hsi}}^{d}, & \text{IoU}\left(B\left(z_{\text{hsi}}^{d}\right), B\left(z_{\text{hsi}}^{1}\right)\right) \geq \rho_b, \\ & s^{\max}\left(z_{\text{hsi}}^{d}\right) - s^{\max}\left(z_{\text{hsi}}^{1}\right) \geq \rho_c, \\ z_{\text{hsi}}^{1}, & \text{otherwise} \end{cases} \tag{17}$$

where B represents the BBOX corresponding to the location with the maximum classification score $s^{\max}$ calculated by $\text{HP}_{w*h*2}^{\text{cls}}$ and z_{hsi}^{1} represents the first-frame template. ρ_b and ρ_c are set to 0.6 and 0.5, respectively. The hyperspectral dynamic template is updated every five frames. When the hyperspectral dynamic template is updated, the corresponding three-band pseudo-color dynamic template (as the SiamBAN's input) is also updated.

E. Differences From the Conference Version of HA-Net

There are four main differences in the journal version of HA-Net compared to the conference version [60].

1) A detailed description of the innovations of HA-Net. HA-Net introduces an online learning module based on the dual Siamese network architecture. A hyperspectral enhancement module is introduced based on the RGB tracker to address the problem of performance limitation of the deep model due to insufficient training data. The online learning module utilizes the first frame to fine-tune the attention layers online to learn features suitable for tracking the current target while updating the dynamic template to adapt to target changes during

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tracking using OC scores. The online learning capability and dynamic template updating strategy of the online learning module can especially improve the generalization ability of the hyperspectral tracker in the case of nonsufficient data.

2) Extensive ablation experiments have been added to validate the validity of the various modules and parameters of the HA-Net. Experiments verify the effectiveness of the dual Siamese network structure. The effectiveness of the dynamic template update strategy and the online fine-tuning of the attention layers are also verified.
3) Comparison experiments with the state-of-the-art hyperspectral/RGB trackers have been added to demonstrate that HA-Net is still competitive. Compared with the state-of-the-art hyperspectral trackers such as SST-Net, TMT-Net, SSTFT, SEE-Net, and SiamHYPER, HA-Net has the best overall performance. Compared to the best RGB trackers such as MixFormer, TransT, and OSTrack, HA-Net also gets the best performance.
4) The performance of HA-Net with fewer training data is verified. We used 25%, 50%, 75%, and 100% of the WHISPERS training data to train HA-Net, and the experiments fully demonstrated the effectiveness of HA-Net. Based on the currently tracked target, tailored online learning fine-tunes the FE model to obtain strongly discriminative features that distinguish the target from the background, thus compensating to some extent for the negative effects of insufficient data.

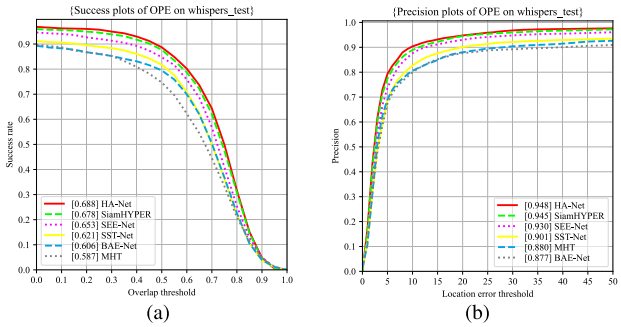
IV. Experimental Results and Analysis

A. Implementation Details

In the experiments conducted in this article, the proposed HA-Net method was implemented in Python with PyTorch 1.6 on an NVIDIA TITAN X (Pascal) GPU. The size of the input area of the pse-template/hyper-template branch of the HA-Net was set to 127×127, and the size of the input area of the HA-Net pse-search/hyper-template branch was set to 255×255. The 1st band, 9th band, and 16th band in the HSI were used to synthesize the three-band PCI. The temporal–spectral–spatial online discrimination learning module did not participate in the offline training and only participated in the online training in the test phase.

1) Training Details: In the training phase, the SiamBAN is the off-the-shelf model. At the same time, all the parameters of the SiamBAN were frozen. During the training process, the Siamese spectral enhancement tracker module was trained from scratch. The Siamese spectral enhancement tracker module was trained for 20 epochs using SGD with an initial learning rate of 0.001. Starting from the first epoch, all the parameters of the Siamese spectral enhancement tracker module were trained. We used approximately 13 000 frames of the training dataset provided as part of the HOT held by IEEE WHISPERS to train HA-Net.

2) Test Details: In the test phase, the *hyper-reg* map of the Siamese spectral enhancement tracker module was removed, leaving only the *hyper-class* map. The classification score of the target was calculated by the hybrid classification feature map $\mathrm{HP}^{\mathrm{cls}}_{w*h*2}$, and the BBOX of the target in the hyperspectral search image was calculated by the *pse-reg* map $P^{\mathrm{reg_all}}_{w*h*4}$ output by the pseudo-color-based tracker. The final classification score was generated by fusing the offline classification score and the OC score output by the temporal–spectral–spatial online discrimination learning module.



(a) (b)

Fig. 3. Results of the hyperspectral object trackers on the WHISPERS dataset. (a) Success plots. (b) Precision plots.

B. Test Dataset and Evaluation Metrics

The dataset provided as part of the WHISPERS comes from the work of Xiong et al. [6]. This dataset provides 40 sequences (about 13 000 frames) for training and 35 sequences (about 17 000 frames) for testing. The dataset contains three types of data: hyperspectral videos (16 bands from 470 to 620 nm), false-color videos (Xiong et al. [6] used the CIE[2] to convert the HSI into false-color image), and RGB videos. The false-color videos were converted from the hyperspectral videos, so the false-color videos and hyperspectral videos are strictly spatially aligned. These videos contain 11 challenging attributes: in-plane rotation (IPR), out of view (OV), motion blur (MB), fast motion (FM), deformation (DEF), out-of-plane rotation (OPR), scale variation (SV), occlusion (OCC), low resolution (LR), illumination variation (IV), and BC [61]. The methods used here as the evaluation metrics are success plots, precision plots, the area under the curve (AUC), and the average distance precision (DP) [62].

C. Comparison of HA-Net and Hyperspectral Trackers

The most advanced hyperspectral trackers [6], [32], [33], [35], [36] in recent years and the HA-Net proposed in this article were compared to verify the advanced performance of HA-Net. All the hyperspectral object trackers were tested on hyperspectral video. The precision plots and the success plots in one-pass evaluation (OPE) for each hyperspectral tracker were then generated. As shown in Fig. 3, HA-Net outperforms all the current state-of-the-art hyperspectral trackers, such as BAE-Net, MHT, SEE-Net, SiamHYPER, and the SST-Net method. Compared with the current best hyperspectral tracker, i.e., SiamHYPER, the proposed HA-Net framework improves the scores by 1% and 0.3%, respectively, for AUC and DP. As shown in Fig. 4, the positioning accuracy and BBOX ratio of the other five hyperspectral trackers in these four

[2]http://cvrl.ioo.ucl.ac.uk/cmfs.htm

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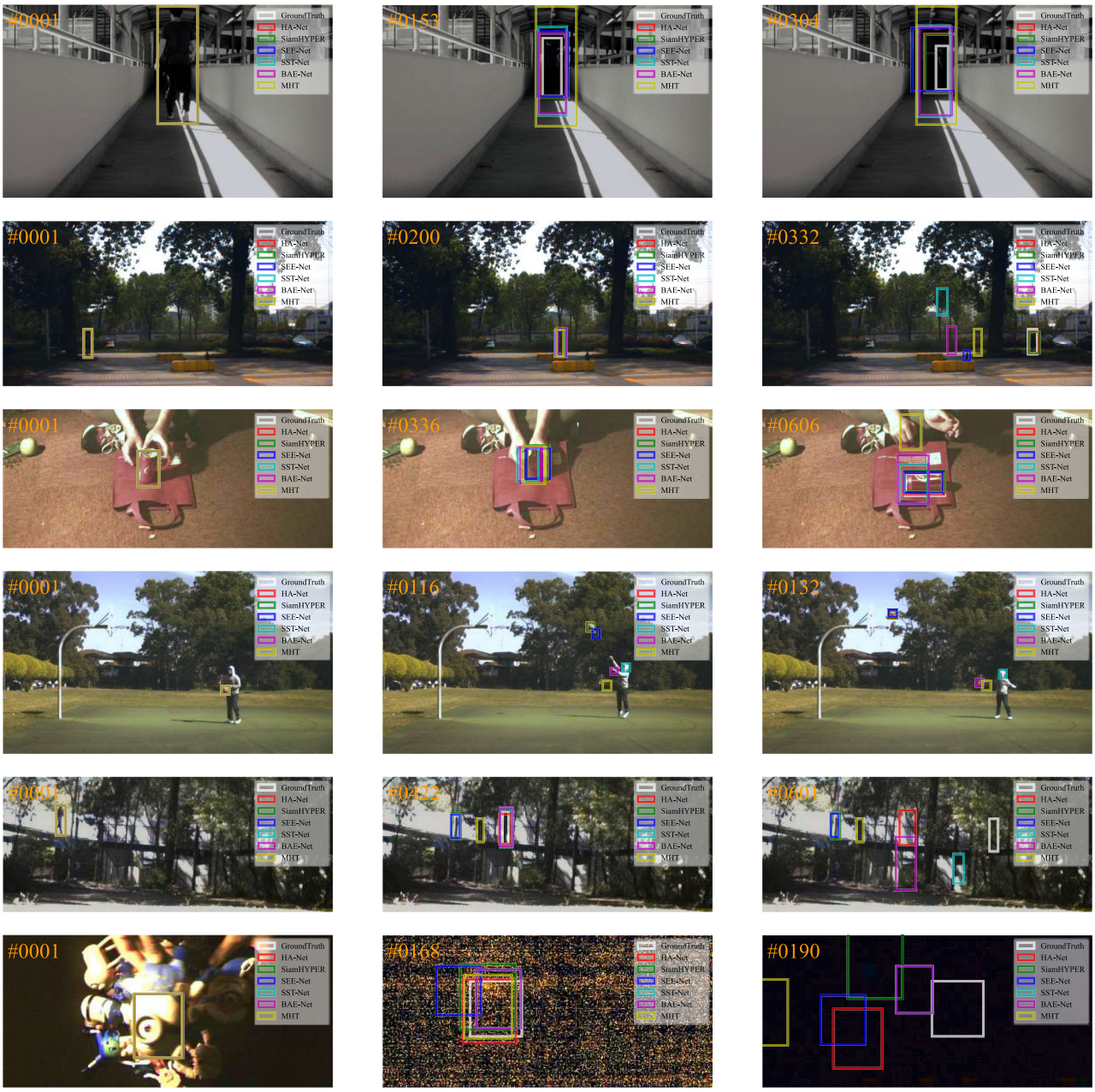


Fig. 4. Comparison of visualization results of the hyperspectral object trackers (scenes: student, pedestrian2, coke, basketball, campus, and toy2).

scenarios (student, pedestrian2, coke, and basketball) are not as good as the positioning accuracy and bounding box ratio of HA-Net. The other hyperspectral trackers lose the tracking target in the pedestrian2 sequences, but SiamHYPER and the proposed HA-Net do not suffer from tracking drift. In the basketball scenario, the basketball is a minute target with limited feature information, and only SEE-Net, SiamHYPER, and HA-Net manage to avoid tracking drift. The basketball and pedestrian2 sequences contain challenging attributes, such as the BC attribute, and the trackers can very easily lose the target under these challenges. This is attributed to the incapacity of existing hyperspectral trackers to adjust to intricate variations in targets, unlike the proposed framework, which can dynamically learn online to accommodate shifts in target complexity even with limited training data. Tracking drift manifested in all hyperspectral trackers within the campus and toy2 scenes. In the campus scene, the trackers faced difficulties, including a similar spectral curve profile trend and variations in illumination (IV), leading to target loss. In the toy2 scene, the hyperspectral trackers lost their targets when light conditions changed. As a result, the BBOX obtained by the proposed HA-Net method is the closest to the ground truth.

Table I compares the running speed of HA-Net with that of the other hyperspectral trackers, such as BAE-Net [32], MHT [6], TMT-Net [63], SSTFT [64], SEE-Net [36], SiamHYPER [35], and SST-Net [33]. BAE-Net, SST-Net, SSTFT, MHT, and HA-Net are hyperspectral trackers with online update modules. In the prediction phase, BAE-Net and SST-Net use multiple weak trackers [32] to calculate the state of the tracking target on the same frame multiple times and then

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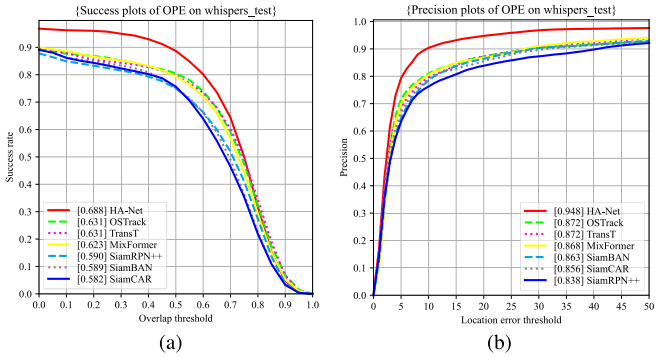
TABLE I
FPS, QUANTITATIVE RESULTS, AND DEVICES OF THE ADVANCED HYPERSPECTRAL TRACKERS WITH THE WHISPERS DATASET

	BAE-Net	MHT	SST-Net	TMT-Net	SSTFT	SEE-Net	SiamHYPER	HA-Net
AUC/DP	0.606/0.877	0.587/0.880	0.621/0.901	0.699/0.928	0.682/0.884	0.653/0.930	0.678/0.945	0.688/0.948
FPS	0.5	0.5	0.45	12.6	19.9	8.72	19	14
Device	GPU: NVIDIA TITAN X (Pascal)	CPU: Intel Core i7 2.8 GHz	GPU: NVIDIA TITAN X (Pascal)	GPU: NVIDIA RTX 3090	GPU: NVIDIA RTX 2080Ti	GPU: NVIDIA TITAN X (Pascal)	GPU: NVIDIA TITAN X (Pascal)	GPU: NVIDIA TITAN X (Pascal)

merge these results to obtain the final state of the target. BAE-Net and SST-Net generate multiple training samples based on the prediction results of each frame. They employ SGD iterations to optimize both the generator and discriminator in the online update process [34]. As a result, BAE-Net and SST-Net can only run at 0.5° and 0.45° frames/s on a GPU, respectively. SSTFT, on the other hand, incorporates a transformer-based context fusion module, enabling efficient utilization of contextual relationships in hyperspectral videos. It also employs an adaptive dynamic template update strategy, yielding commendable performance. SSTFT can run at 19.9° frames/s on the NVIDIA RTX 2080Ti. However, SSTFT employs a linear template updating strategy that lacks online learning capability. This approach poses challenges in effectively capturing the dynamics of the target, particularly in scenarios with limited training data. Although MHT obtains good results, MHT combines SSHMG [6] and the hyperspectral abundance as a hybrid feature, which requires a large amount of calculation, so it only runs at 0.5° frames/s on a CPU. The linear update strategy employed in MHT is susceptible to error accumulation, resulting in tracking drift. TMT-Net, despite achieving high accuracy through three Siamese networks for FE, introduces redundant parameters. In addition, TMT-Net incorporates a fine-tuning module with a Siamese network, reducing running speed. SEE-Net, while enhancing tracking results with a spectral self-expression module for generating crucial PCIs, experiences a reduction in running speed. In addition to improving the accuracy, the proposed framework can run at 14° frames/s on a GPU.

D. Comparison of HA-Net and RGB-Based Siamese Tracker

HA-Net was experimented on hyperspectral videos, and SiamCAR [17], SiamRPN++ [16], SiamBAN [44], OSTrack [50], TransT [65], and MixFormer [49] were experimented on the false-color videos. These approaches are currently advanced RGB-based object tracking approaches. The success plots and the precision plots in OPE for each tracker were then generated. As shown in Fig. 5, compared with the anchor-free RGB-based deep tracker, i.e., SiamBAN, HA-Net obtains an AUC score of 0.688 and a DP score of 0.948, representing improvements of 9.9% and 8.5%, respectively, for AUC and DP. Compared with the anchor-based RGB-based deep tracker, i.e., SiamRPN++, HA-Net improves the scores by 9.9% and 11.0%, respectively, for AUC and DP. OSTrack, TransT, and MixFormer represent transformer-based RGB trackers, outperforming traditional pure CNN-based RGB trackers and showcasing the potential of transformers in single object tracking. MixFormer, in particular, incorporates an online score prediction module for template updates, yielding commendable tracking results. However, the absence of online learning capabilities imposes limitations on MixFormer's overall performance. Since hyperspectral video has richer spectral information than RGB video, allowing the identification of objects based on the material of the objects, by using hyperspectral features, the proposed framework outperforms the RGB-based trackers.



(a) (b)

Fig. 5. Quantitative results of HA-Net and RGB-based trackers on the WHISPERS dataset. (a) Success plots. (b) Precision plots.

As shown in Table II, SiamCAR, SiamRPN++, SiamBAN, OSTrack, TransT, and MixFormer were tested on false-color videos and RGB videos. HA-Net, SiamHYPER, SEE-Net, and SST-Net were tested on hyperspectral videos. For all these trackers, HA-Net obtains the best results.

E. Attribute-Based Comparison

All different types of object trackers were experimented on corresponding false-color videos and hyperspectral videos. The precision plots and the success plots in OPE for each tracker were then generated. Fig. 6 shows the success plots and precision plots for the BC attribute. The proposed HA-Net framework obtains an AUC score of 0.727 for the BC attribute and a DP score of 0.981 for the BC attribute. In the context of the BC attribute, HA-Net exhibits improvements of 2.5% and 2.1%, respectively, for AUC and DP when compared with SiamHYPER. The RGB-based trackers cannot

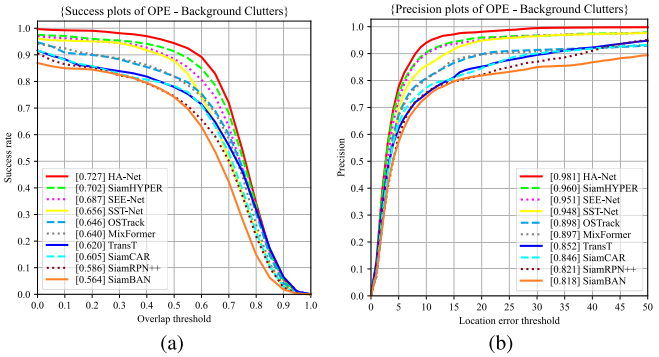
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TABLE II
AUC/DP SCORES OF DIFFERENT TRACKERS ON THE WHISPERS DATASET

	SiamBAN	SiamCAR	MixFormer	TransT	OSTrack	SST-Net	SEE-Net	SiamHYPER	HA-Net
RGB	0.606/0.847	0.599/0.858	0.660/0.915	0.667/0.901	0.671/0.925	a/n	a/n	a/n	a/n
Hyperspectral	a/n	a/n	a/n	a/n	a/n	0.621/0.901	0.653/0.930	0.678/0.945	0.688/0.948
False-color	0.589/0.863	0.582/0.856	0.623/0.868	0.631/0.872	0.631/0.872	a/n	a/n	a/n	a/n

TABLE III
ATTRIBUTE-BASED AUC SCORES ON THE WHISPERS DATASET

Attribute	SiamBAN	SiamCAR	SiamRPN++	MixFormer	TransT	OSTrack	SST-Net	SEE-Net	SiamHYPER	HA-Net
BC	0.564	0.604	0.587	0.640	0.620	0.646	0.656	0.687	0.702	0.727
DEF	0.681	0.664	0.684	0.691	0.702	0.733	0.674	0.699	0.720	0.712
FM	0.635	0.678	0.559	0.633	0.666	0.677	0.583	0.686	0.710	0.713
IPR	0.688	0.653	0.674	0.705	0.742	0.723	0.693	0.723	0.720	0.744
IV	0.515	0.452	0.478	0.597	0.536	0.584	0.407	0.563	0.592	0.608
LR	0.573	0.510	0.475	0.596	0.536	0.553	0.448	0.614	0.664	0.671
MB	0.668	0.656	0.539	0.615	0.666	0.659	0.564	0.683	0.750	0.734
OCC	0.561	0.561	0.544	0.637	0.607	0.627	0.584	0.614	0.635	0.646
OPR	0.693	0.645	0.697	0.689	0.722	0.735	0.689	0.711	0.706	0.730
OV	0.616	0.582	0.608	0.799	0.741	0.806	0.515	0.622	0.596	0.603
SV	0.611	0.548	0.591	0.630	0.619	0.635	0.597	0.639	0.657	0.672



(a) (b)

Fig. 6. Quantitative results of different trackers with attribute-based BC on the WHISPERS dataset. (a) Success plots. (b) Precision plots.

use hyperspectral information, so in the case of the BC attribute, the RGB-based trackers have difficulty in discriminating objects and backgrounds with a similar appearance. Compared with the best-performing RGB-based tracker, i.e., OSTrack, HA-Net shows improvements of 8.1% and 8.3%, respectively, for AUC and DP. Compared with both the existing hyperspectral and RGB-based trackers, HA-Net obtains superior results. As shown in Table III, HA-Net ranks first in IPR, FM, OCC, SV, IV, LR, and BC.

V. ABLATION EXPERIMENTS

We performed ablation experiments on the WHISPERS dataset to prove the effectiveness of every module of the proposed framework. As shown in Table IV, the baseline is SiamBAN, which is an RGB-based anchor-free object tracker. The SiamBAN model is the off-the-shelf model, which was trained with massive RGB data. We tested the baseline on false-color videos and obtained an AUC score of 0.589 and a DP score of 0.863. Based on SiamBAN, we added a Siamese spectral enhancement tracker module trained by the hyperspectral training samples provided by the WHISPERS dataset. During the training and testing phases, we removed the hyperspectral regression branch (SiamBAN+SEM-HR) and obtained an AUC score of 0.651 and a DP score of 0.922. Then, we tested the baseline + Siamese spectral enhancement tracker module (SiamBAN+SEM: the hyperspectral regression branch is included in the training phase and removed in the test phase) on the hyperspectral videos and obtained an AUC score of 0.662 and a DP score of 0.931. Compared with the baseline, the baseline + Siamese spectral enhancement tracker module improves the scores by 7.3% and 6.8%, respectively, for AUC and DP. Finally, we added the temporal–spectral–spatial online discrimination learning module based on the double Siamese hyperspectral object tracking network. Compared with the baseline, HA-Net improves the scores by 9.9% and 8.5%, respectively, for the AUC and DP. If we add a hyperspectral regression map, the effect is not as good as that of HA-Net. This is because the task of regression requires more training samples. The pseudo-color-based tracker was trained using tens of millions of RGB training samples to obtain a stable and robust regression branch. This guarantees the generalization of HA-Net.

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TABLE IV
ABLATION EXPERIMENT OF HA-NET ON THE WHISPERS DATASET

Method	Training sample	Test sample	AUC	ΔAUC	DP	ΔDP
SiamBAN	RGB	False-color	0.589	-	0.863	-
SiamBAN+SEM-HR	Hyperspectral	Hyperspectral	0.651	+6.3%	0.922	+5.9%
SiamBAN+SEM	Hyperspectral	Hyperspectral	0.662	+7.3%	0.931	+6.8%
SiamBAN+SEM+HR+TSS	Hyperspectral	Hyperspectral	0.683	+9.4%	0.942	+7.9%
SiamBAN+SEM+TSS (HA-Net)	Hyperspectral	Hyperspectral	0.688	+9.9%	0.948	+8.5%

• The baseline is SiamBAN. SEM: the Siamese spectral enhancement tracker module; HR: hyperspectral enhancement regression feature map; TSS: temporal-spectral-spatial online discrimination learning module; SEM-HR: remove the hyperspectral regression branch during both training phase and testing phase.

TABLE V
INFLUENCE OF RGB/HYPERSPECTRAL TRAINING SAMPLES ON THE DEEP SIAMESE OBJECT TRACKERS WITH THE WHISPERS DATASET

Method	Training sample	Test sample	AUC	DP
SiamBAN_16	Hyperspectral	Hyperspectral	0.490	0.797
SiamBAN	RGB	False-color	0.589	0.863
HA-Net	Hyperspectral	Hyperspectral	0.688	0.948

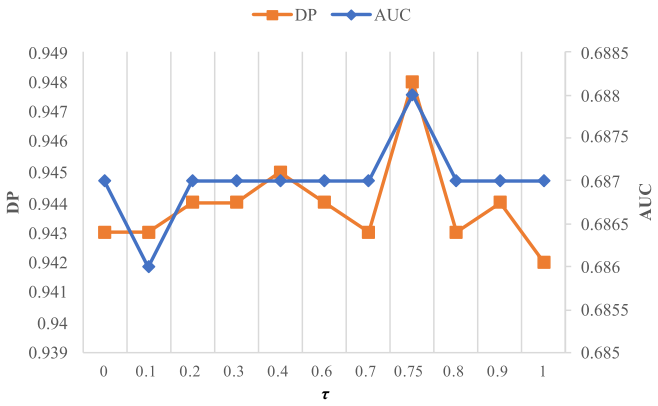


Fig. 7. Effect of the value of τ on the AUC/DP score of HA-Net with the WHISPERS dataset.

We used different datasets to train the baseline to show the influence of various training data on the behavior of the final model. As shown in Table V, based on the official SiamBAN code, we only changed the input channel of SiamBAN to 16 for the training from scratch (no pretrained hyperspectral model) with the hyperspectral training samples (approximately 13 000 frames) provided in the WHISPERS dataset (which is denoted as SiamBAN_16). The model of SiamBAN we used was the off-the-shelf model, which was trained with massive RGB data. SiamBAN_16 was experimented on hyperspectral videos, and SiamBAN was tested on false-color videos. Due to the insufficient training data, the performance of SiamBAN_16 was not as good as that of SiamBAN trained with massive RGB training samples. This is why the existing hyperspectral trackers [6], [22], [23], [32] can only use pseudo-color features or handcrafted features. Therefore, we propose the double Siamese network structure to use multilevel hyperspectral features from deep learning (from shallow information to semantic information), which greatly improves the running speed accuracy and accuracy of the hyperspectral tracker. At the same time, the robustness of HA-Net framework proposed in this article is improved by modeling temporal–spectral–spatial information to prevent model degradation under complex conditions.

As shown in Table VI, we tested the effects of parameters D_T and θ, the FE module, and features of the Siamese spectral enhancement tracker module (SMD)/features of the pseudo-color-based Siamese tracker module (PMD) on the final results. Comparing **EX1** and **EX4**, the PMD obtains better results because the PMD is trained with massive RGB training samples and is more robust than the SMD. Comparing **EX2** with **EX3**, the effectiveness of the FE module is demonstrated without dynamic template updates. Online fine-tuning of the FE module can make HA-Net adapt to the emergence of new targets and the changes of targets during tracking and reduce the risk of HA-Net overfitting. In **EX2**, the effect of different values of θ on the results is shown, and the effect of θ on the results is relatively large, compared to the other parameters. When the value of θ is 0.7 or 0.8, HA-Net obtains the top two results. In **EX4**, $\theta = 0.7$ or $\theta = 0.8$ was chosen to test the effect of updating the dynamic template on the tracking results. It can be concluded from **EX4** that HA-Net obtains the best results when the dynamic template is updated with a period of $D_T = 5$ and $\theta = 0.8$.

As shown in Fig. 7, we tested different values of τ on the WHISPERS dataset, and HA-Net obtained the best AUC/DP score when $\tau = 0.75$. As shown in Fig. 8, HA-Net achieves the best AUC/DP score when $\rho_b = 0.6$ and $\rho_c = 0.5$. We found that ρ_b and ρ_c do not have a significant impact on the final results of the WHISPERS dataset, but we still experimentally verified that the values we chose for these two parameters promote HA-Net to obtain the best results.

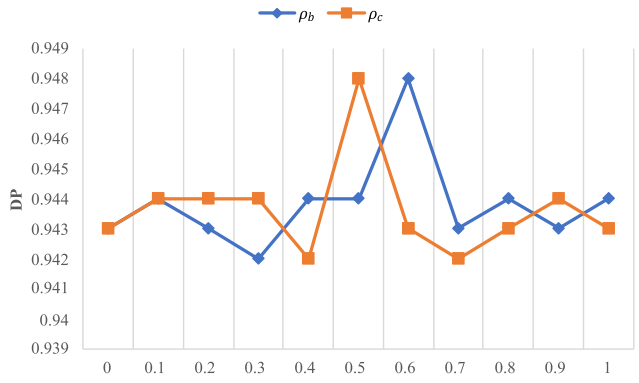
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TABLE VI

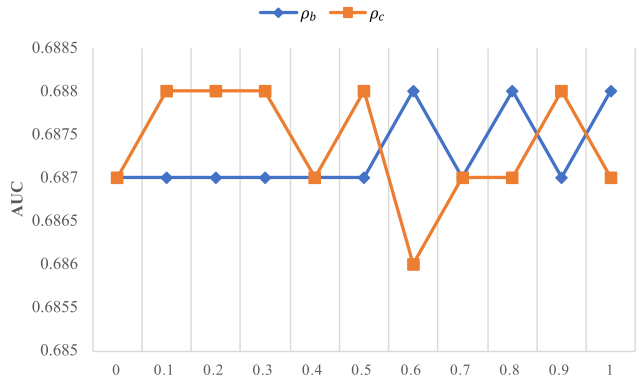
ABLATION EXPERIMENTS WITH DIFFERENT MODELS AND PARAMETERS D_T AND θ WITH THE WHISPERS DATASET

EXP	D_T	θ	τ	ρ_b	ρ_c	FE	SMD/PMD	AUC/DP	EXP	D_T	θ	τ	ρ_b	ρ_c	FE	SMD/PMD	AUC/DP
EX1	5	0.8	0.75	0.6	0.5	✓	SMD	**0.685/0.944**	EX3	-	0.7	0.75	0.6	0.5	-	PMD	0.670/**0.930**
	-	0.0	0.75	0.6	0.5	✓	PMD	0.630/0.884		-	0.8	0.75	0.6	0.5	-	PMD	**0.673**/0.924
	-	0.1	0.75	0.6	0.5	✓	PMD	0.642/0.890	EX4	1	0.7	0.75	0.6	0.5	✓	PMD	0.686/0.943
	-	0.2	0.75	0.6	0.5	✓	PMD	0.665/0.921		5	0.7	0.75	0.6	0.5	✓	PMD	0.685/0.941
	-	0.3	0.75	0.6	0.5	✓	PMD	0.682/0.938		10	0.7	0.75	0.6	0.5	✓	PMD	0.686/0.944
	-	0.4	0.75	0.6	0.5	✓	PMD	0.668/0.921		15	0.7	0.75	0.6	0.5	✓	PMD	0.686/0.942
EX2	-	0.5	0.75	0.6	0.5	✓	PMD	0.682/0.939		20	0.7	0.75	0.6	0.5	✓	PMD	0.686/0.944
	-	0.6	0.75	0.6	0.5	✓	PMD	0.674/0.926		1	0.8	0.75	0.6	0.5	✓	PMD	0.686/0.943
	-	0.7	0.75	0.6	0.5	✓	PMD	0.680/**0.941**		5	0.8	0.75	0.6	0.5	✓	PMD	**0.688/0.948**
	-	0.8	0.75	0.6	0.5	✓	PMD	**0.683**/0.939		10	0.8	0.75	0.6	0.5	✓	PMD	0.688/0.943
	-	0.9	0.75	0.6	0.5	✓	PMD	0.679/0.933		15	0.8	0.75	0.6	0.5	✓	PMD	0.687/0.944
	-	1.0	0.75	0.6	0.5	✓	PMD	0.660/0.904		20	0.8	0.75	0.6	0.5	✓	PMD	0.686/0.942

• D_T: update cycle of dynamic templates; θ: fusion weights of online classification scores and offline classification scores; FE: feature extraction module; SMD/PMD: features of the Siamese spectral enhancement tracker module/features of the pseudo-color based Siamese tracker module.



(a)



(b)

Fig. 8. AUC/DP score obtained for different values of ρ_b and ρ_c with the WHISPERS dataset. (a) DP score obtained for different values of ρ_b and ρ_c with the WHISPERS dataset. (b) AUC score obtained for different values of ρ_b and ρ_c with the WHISPERS dataset.

VI. DATA SCALE ABLATION EXPERIMENTS

The scarcity of training samples will persist as a major constraint on the performance of hyperspectral trackers for the foreseeable future, primarily due to the challenges associated with acquiring hyperspectral video data. Consequently, the capability of a hyperspectral tracker to engage in online learning within limited training samples proves instrumental in enhancing model generalization, particularly in complex scenarios.

TABLE VII

EVALUATION OF TRAINING HA-NET ON DIFFERENT DATA SCALES

AUC / Data Scale / Method	25%	50%	75%	100%
SiamBAN+SEM	0.591	0.648	0.656	0.662
SiamBAN+SEM+TSS (HA-Net)	0.668	0.671	0.680	0.688

• The baseline is SiamBAN. SEM: the Siamese spectral enhancement tracker module; TSS: temporal-spectral-spatial online discrimination learning module.

As shown in Table VII, we conducted experiments to train HA-Net using varying dataset scales to assess its performance under the constraint of limited training data. SiamBAN+SEM and SiamBAN+SEM+TSS (HA-Net) were trained using 25%, 50%, 75%, and 100% of the training data from the WHISPERS dataset. For SiamBAN+SEM, AUC scores of 0.591, 0.648, 0.656, and 0.662 were achieved with 25%, 50%, 75%, and 100% of the training data, respectively. The experiments demonstrated the effectiveness of pseudo-color and hyperspectral fusion in mitigating the challenges posed by limited hyperspectral video tracking training data. For SiamBAN+SEM+TSS (HA-Net), AUC scores of 0.668, 0.671, 0.680, and 0.688 were obtained with 25%, 50%, 75%, and 100% of the training data, respectively. Notably, HA-Net trained with 25% of the training data exhibited a 7.7% improvement in AUC compared to SiamBAN+SEM. The results underscore the excellent tracking performance of HA-Net in complex scenarios with limitations in training data, attributed to the benefits of online learning.

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VII. Conclusion

In this article, a deep temporal–spectral–spatial anchor-free Siamese tracking network for hyperspectral video object tracking, namely, HA-Net, has been proposed for hyperspectral video object tracking. The double Siamese network is trained in an end-to-end manner and can use deep hyperspectral semantic features for the insufficient training data. HA-Net can directly predict the hyperspectral target category and regress the bounding boxes, without multiscale searching or candidate boxes. The proposed architecture also integrates a temporal–spectral–spatial online discrimination learning module to enhance the ability of the double Siamese network to deal with distractors. Simultaneously, the temporal–spectral–spatial online discrimination learning module offers a robust online learning capability, enhancing the robustness and accuracy of HA-Net in complex scenarios. This module effectively learns the target's changes online, even within the constraints of limited training data, thereby improving the model's generalization and performance. The framework we have proposed is simple and could be easily improved in the future. The experiments conducted in this article proved that the proposed HA-Net framework can obtain the best results on the 2021 IEEE WHISPERS Hyperspectral Object Tracking (HOT) dataset. At the same time, HA-Net can operate at 14 frames/s on a GPU, which is much faster than the current hyperspectral object trackers.

Acknowledgment

The authors would like to thank the editor, the associate editor, and anonymous reviewers for their helpful comments and suggestions. They would also like to give special thanks to Fengchao Xiong, Jun Zhou, Zhuanfeng Li, and Yuntao Qian for sharing the dataset, and the authors of the algorithms used in the experiments for sharing their codes.

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